



BIO262 PHYSIOLOGY

MODULE 5

MUSCLES

NEUROMUSCULAR JUNCTIONS

NERVES

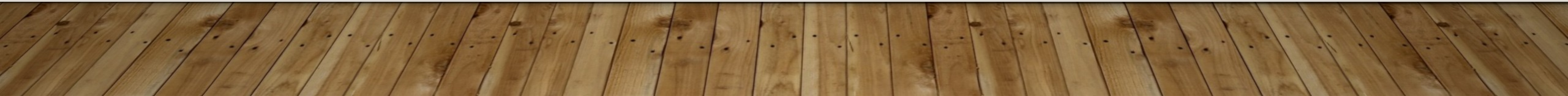
GENERAL MUSCLE INFO:

Name	Vol or not	Striated or not	Nuclei/cell	Repair or not	Where?
Skeletal	Voluntary	Yes	Hundreds	NO - scars	Covers bones
Cardiac	Non	Yes	2-5	NO - scars	Heart - blood through body
Smooth	Non	No	ONE	YES	Vessels, hollow organs, eyes....

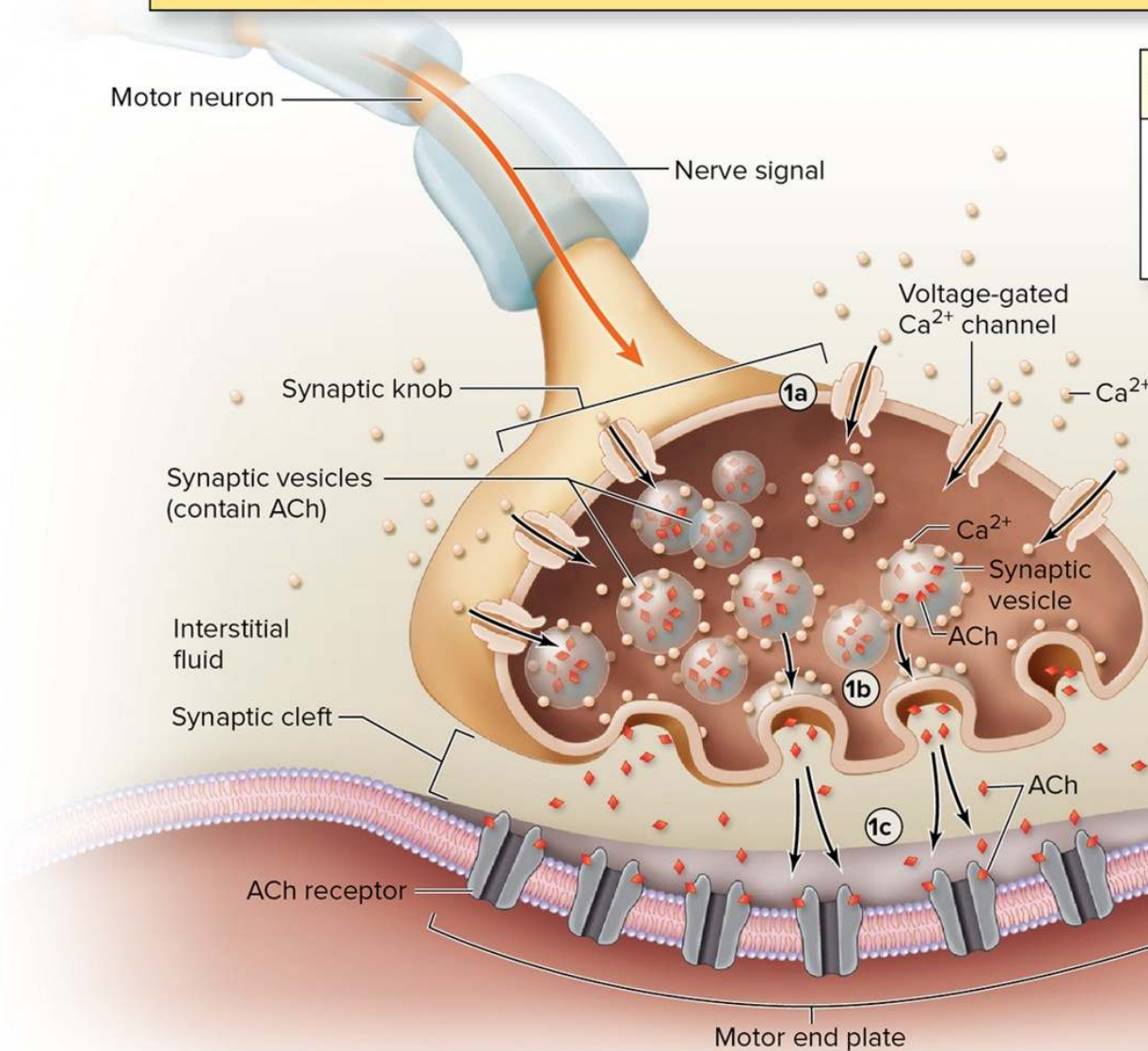
FUNCTIONS:

- ❖ **Movement (skeletal - movement, balance), (Smooth - moves stuff through GI, GU tracts, vessel diameter)**
- ❖ Posture - TONIC stimulus - never stops working
- ❖ Supports organs - pelvic floor, abdominal wall
- ❖ Stabilize joints - strong muscles protect joints
- ❖ Guards opening - sphincters!
- ❖ Generates heat
- ❖ Stores nutrients - Glycogen, myoglobin

Cardiac - Very resistant to fatigue due to high number of mitochondria.



1 NEUROMUSCULAR JUNCTION: EXCITATION OF A SKELETAL MUSCLE FIBER



1a Ca^{2+} entry at synaptic knob

A nerve signal is propagated down a motor axon and triggers the entry of Ca^{2+} into the synaptic knob.

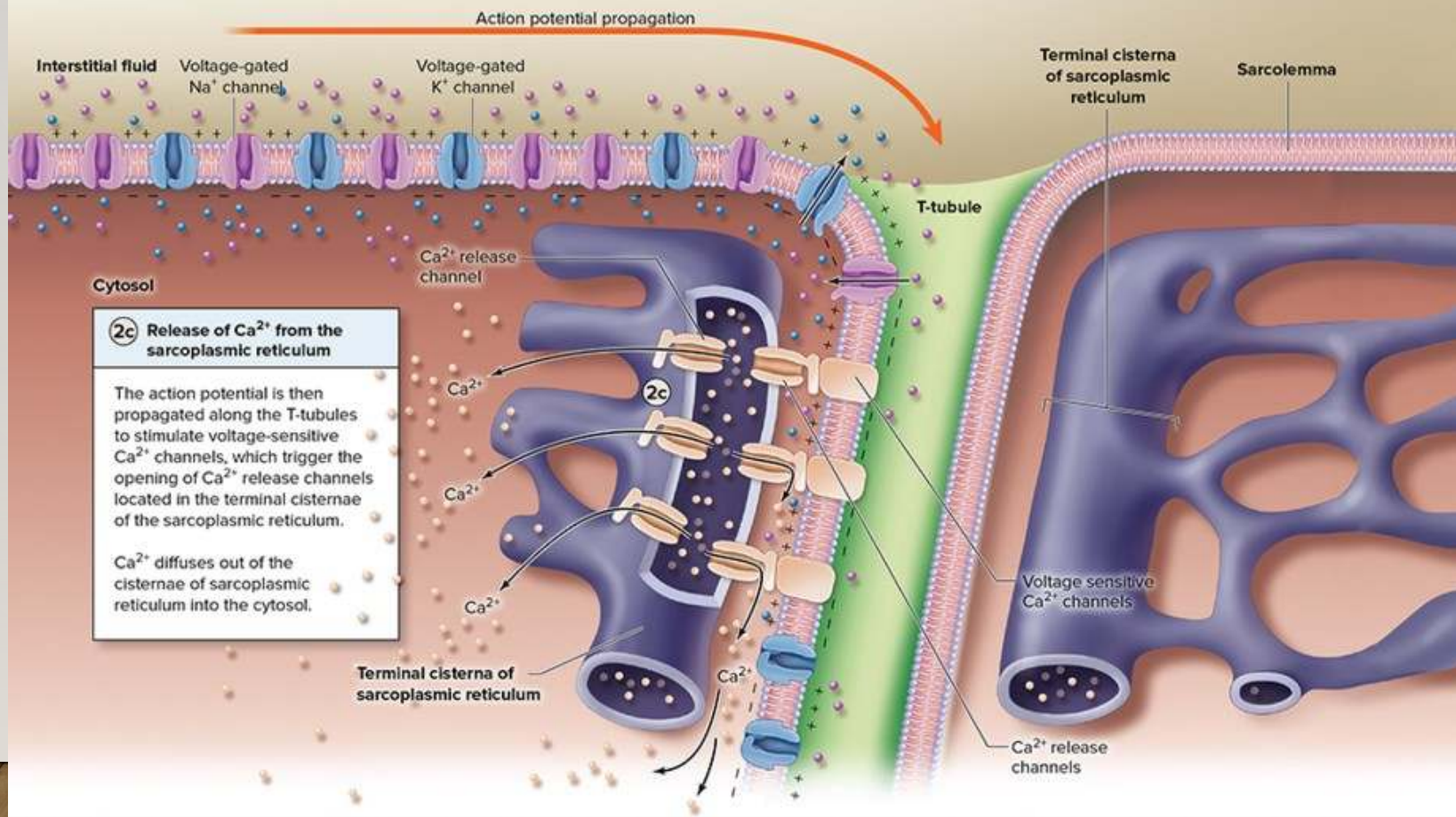
Ca^{2+} binds to proteins in synaptic vesicle membrane.

1b Release of ACh from synaptic knob

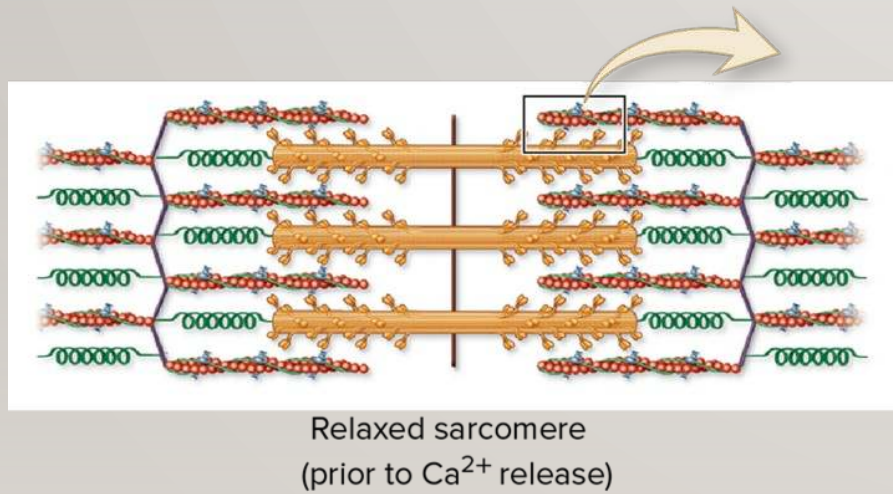
Calcium binding triggers synaptic vesicles to merge with the synaptic knob plasma membrane and ACh is exocytosed into the synaptic cleft.

1c Binding of ACh to ACh receptor at motor end plate

ACh diffuses across the fluid-filled synaptic cleft at the motor end plate to bind with ACh receptors.

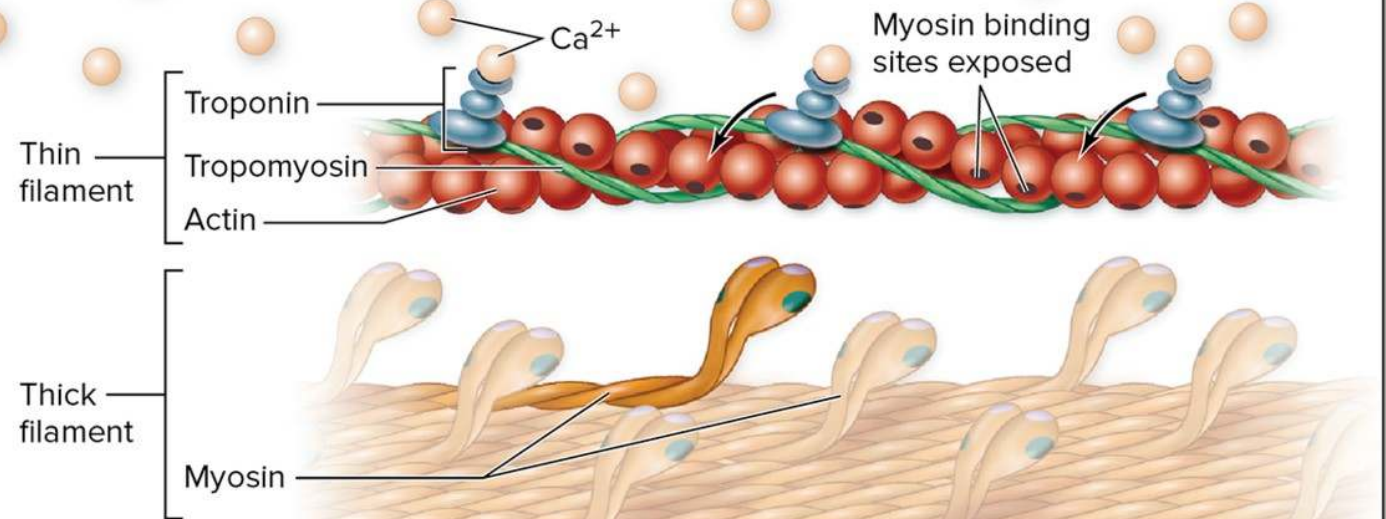


3 SARCOMERE: CROSSBRIDGE CYCLING



3a Ca^{2+} binding

Ca^{2+} binds to troponin in muscle thin filaments, causing a conformational change in troponin. Troponin changes shape and the entire troponin-tropomyosin complex is moved—thus, tropomyosin no longer covers the myosin binding site on actin.



1 NEUROMUSCULAR JUNCTION: EXCITATION OF A SKELETAL MUSCLE FIBER

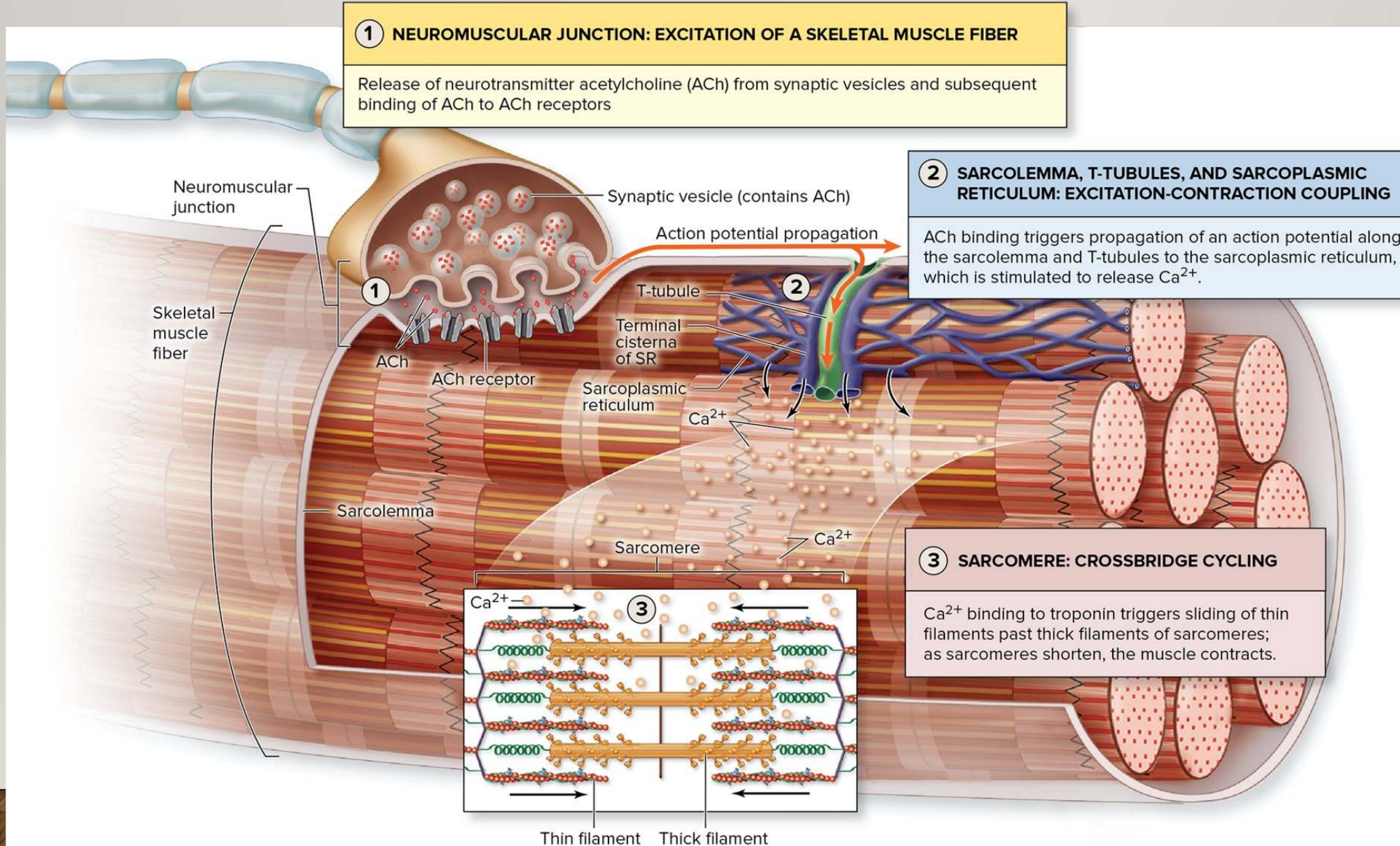
Release of neurotransmitter acetylcholine (ACh) from synaptic vesicles and subsequent binding of ACh to ACh receptors

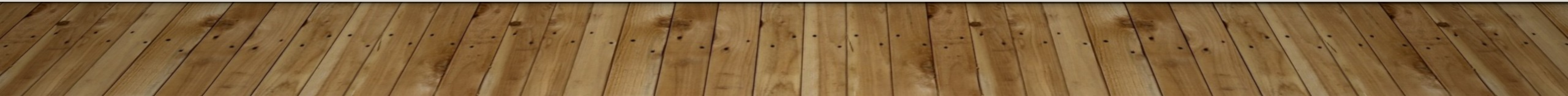
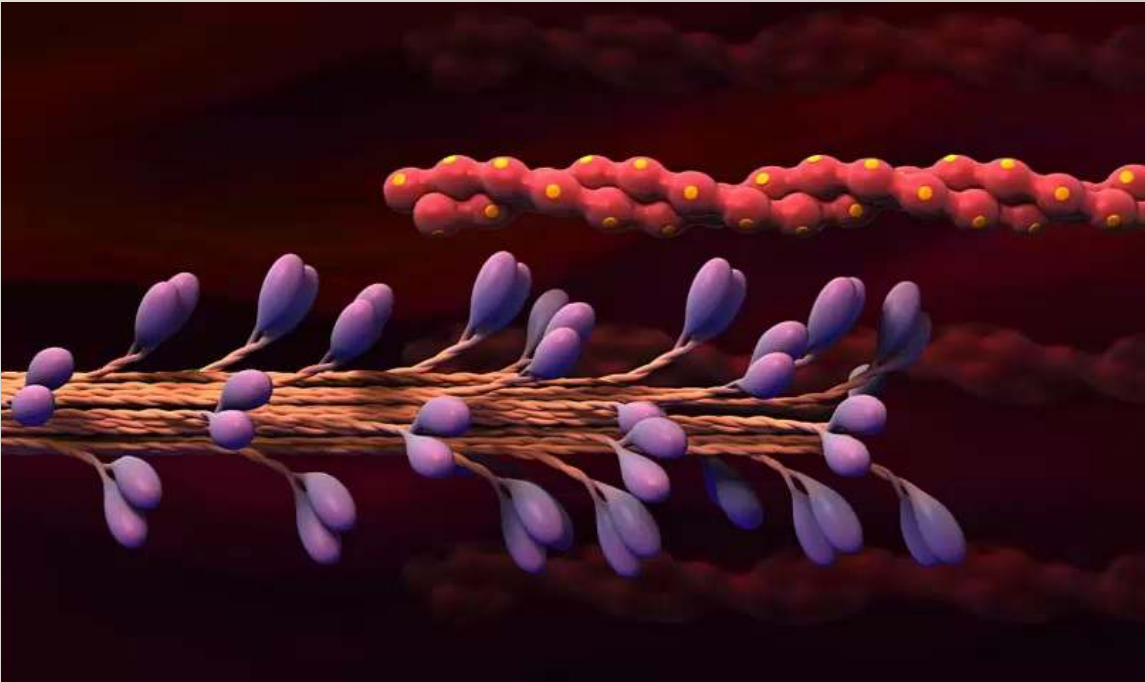
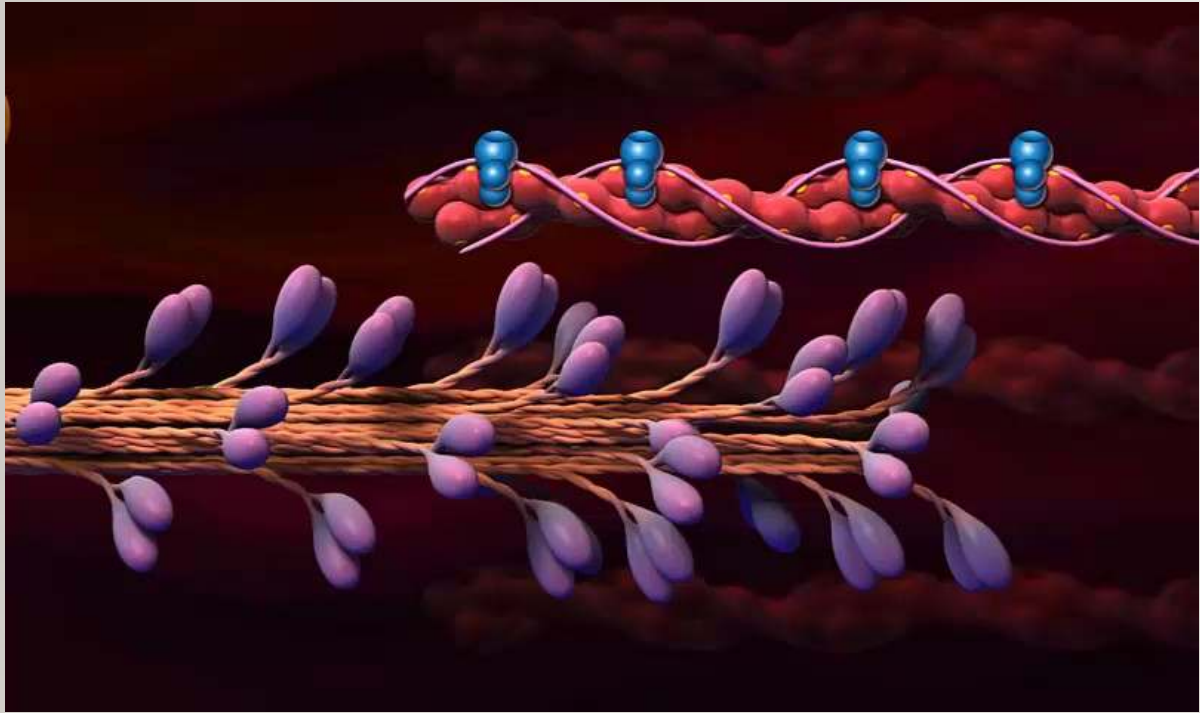
2 SARCOLEMMMA, T-TUBULES, AND SARCOPLASMIC RETICULUM: EXCITATION-CONTRACTION COUPLING

ACh binding triggers propagation of an action potential along the sarcolemma and T-tubules to the sarcoplasmic reticulum, which is stimulated to release Ca^{2+} .

3 SARCOMERE: CROSSBRIDGE CYCLING

Ca^{2+} binding to troponin triggers sliding of thin filaments past thick filaments of sarcomeres; as sarcomeres shorten, the muscle contracts.





PRACTICE QUESTION

- The absence of a sodium ion influx would mean?

A – extra calcium is released from the terminal cisternae

B – the troponin/tropomyosin complex does not move

C – a successful contraction will occur

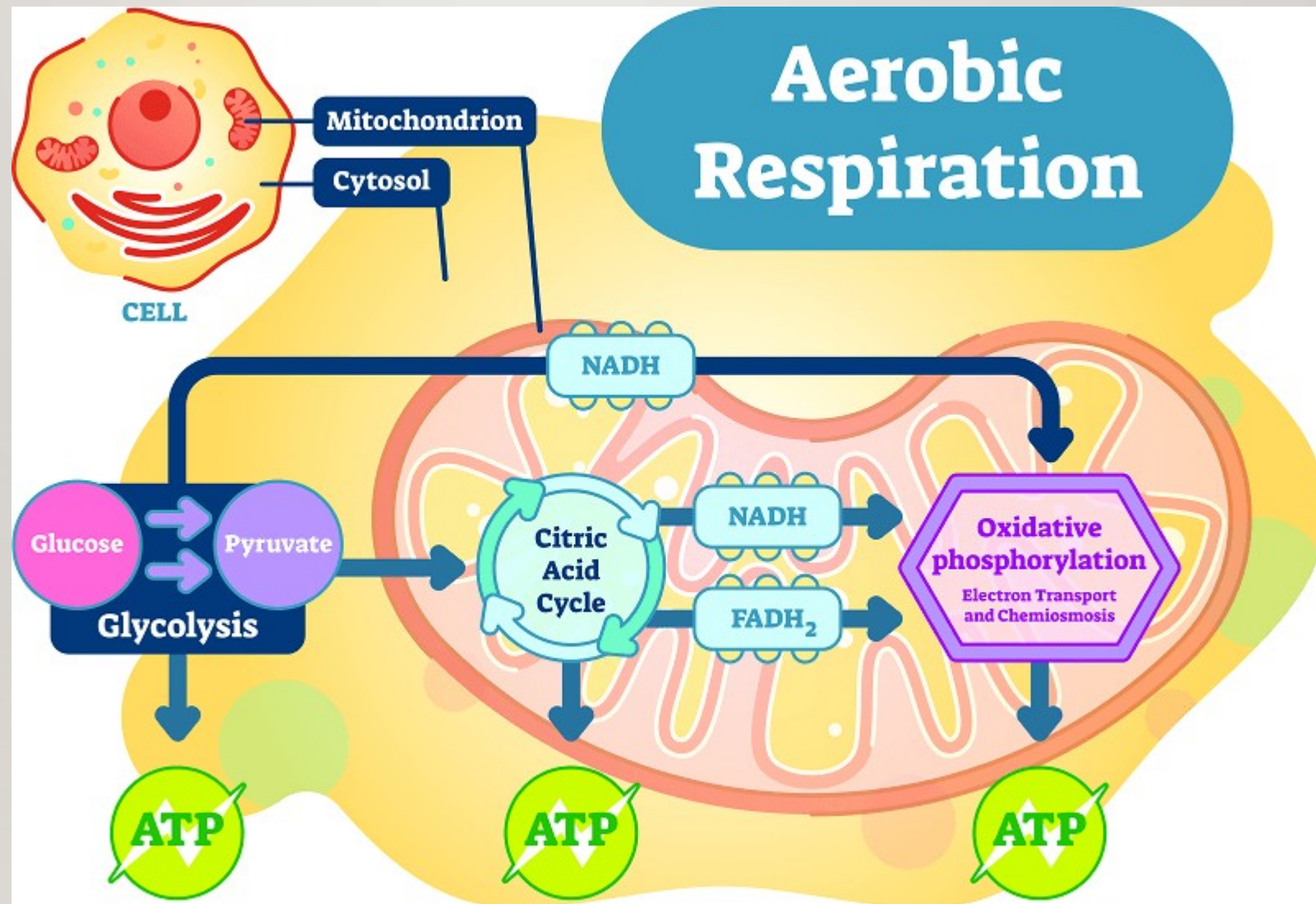
D – ATP production will not occur

MITOCHONDRIA- MAKING ATP ("BATTERY" FULL OF ENERGY)

- **Glycolysis**: one glucose breaks into two pyruvate (**NO O₂** needed- **anaerobic**)
 - Happens in the **cytoplasm** – no organelle needed – **only 2 ATP made**
- Pyruvate moves into the mitochondria (**MUST have O₂ - aerobic**)
- CO₂ is expelled which refines the pyruvate (makes it reactive)
- This moves into the **Krebs/Citric acid cycle** (a series of chemical reactions – inside the **mitochondria** matrix)
- H⁺ and O₂ combine to make water ** Oxygen is needed**
- AND the transfer of H⁺ phosphorylates ADP to make ATP – Electron Transport chain in the membrane of the mitochondria.
- *****34 ATP is made in the processes inside the mitochondria (aerobic)!*******

What happens to pyruvate if NO oxygen?

Anaero
bic



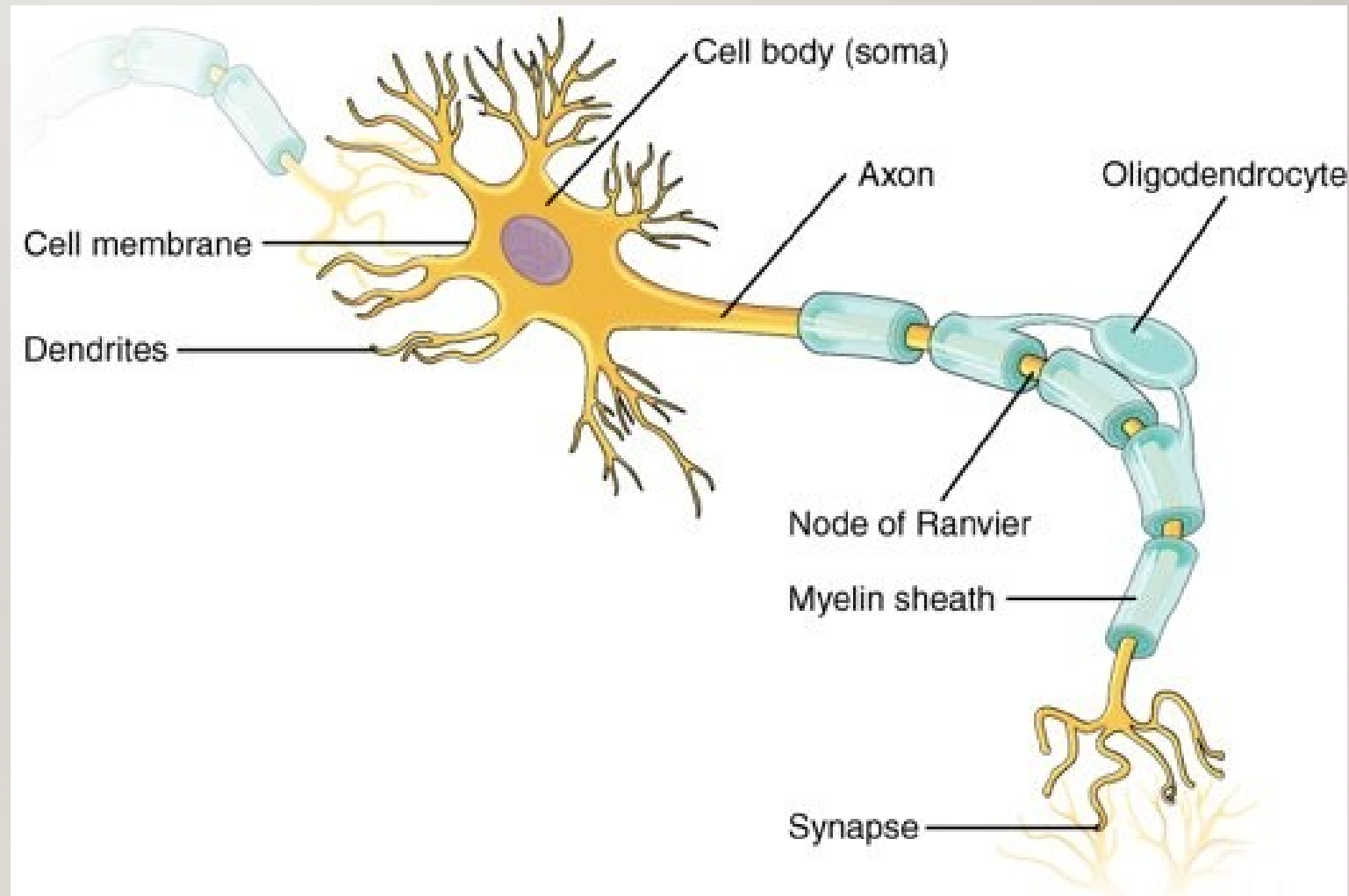
NEURON STRUCTURE AND FUNCTION:

Remember! This is a cell, so it has all the organelles in it that we already learned!

MAIN Function:

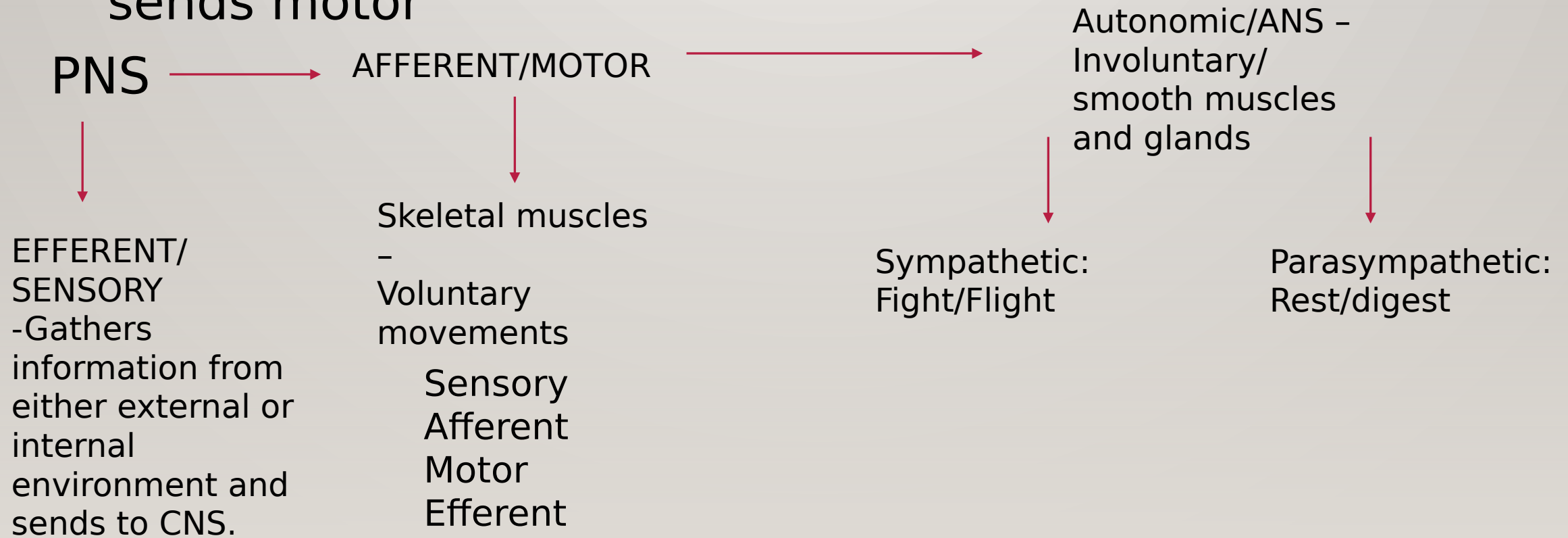
Transmission of electrical signals:

- Sensory to CNS (brain/spinal cord)
- Motor from CNS to muscles/glands



DIVISIONS OF THE NERVOUS SYSTEM AND FUNCTIONS (DIRECTION)

CNS= Brain and spinal cord – processes sensory and sends motor

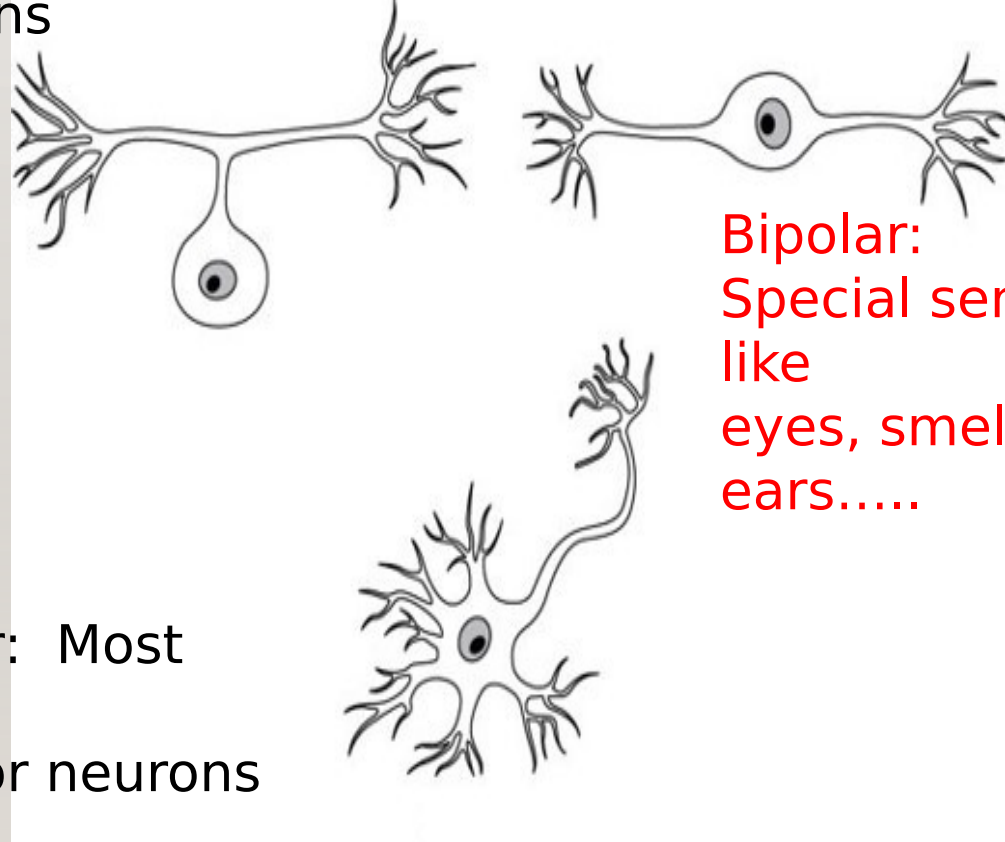


CLASSIFICATION OF NEURONS - STRUCTURAL

Anaxonic
:
Found in
brain
and
some
special
senses
(fairly
rare)



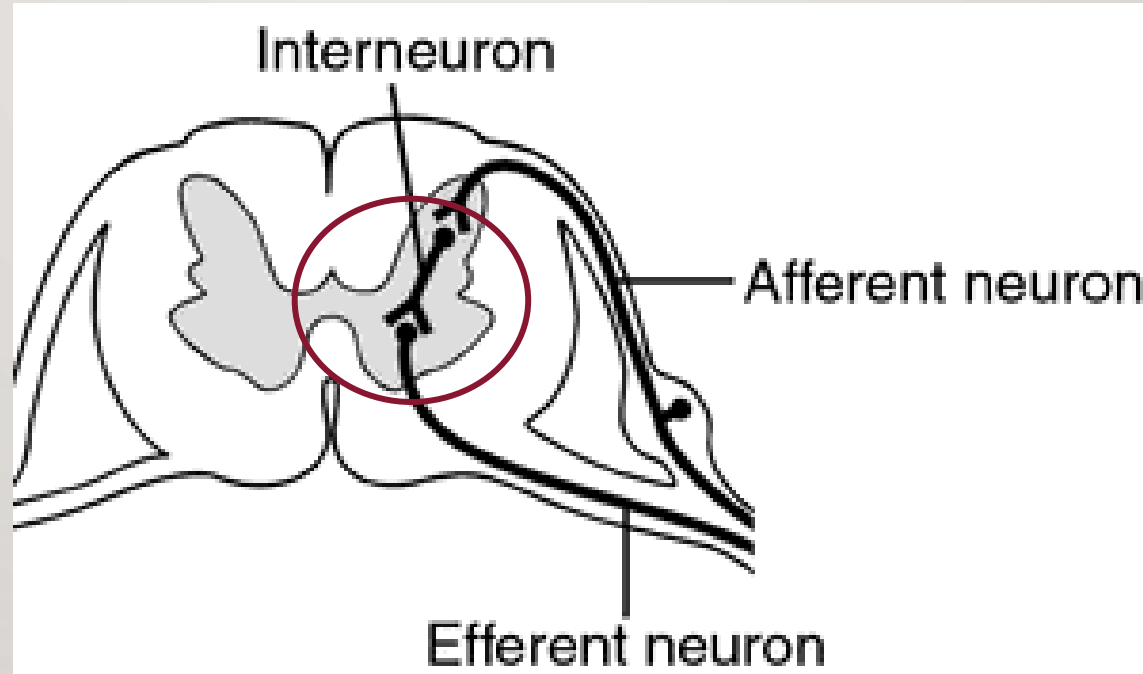
Unipolar:
SENSORY
neurons



Bipolar:
Special senses
like
eyes, smell,
ears.....

Multipolar: Most
common
ALL motor neurons

**Interneurons connect
sensory to motor
within the spinal cord.
Facilitates spinal
reflexes!**



Just some fun
videos
showing
neurons
firing.....

<https://www.youtube.com/watch?v=j7aOwjGLOq0>

<https://www.youtube.com/watch?v=lhkK6jURLjs>

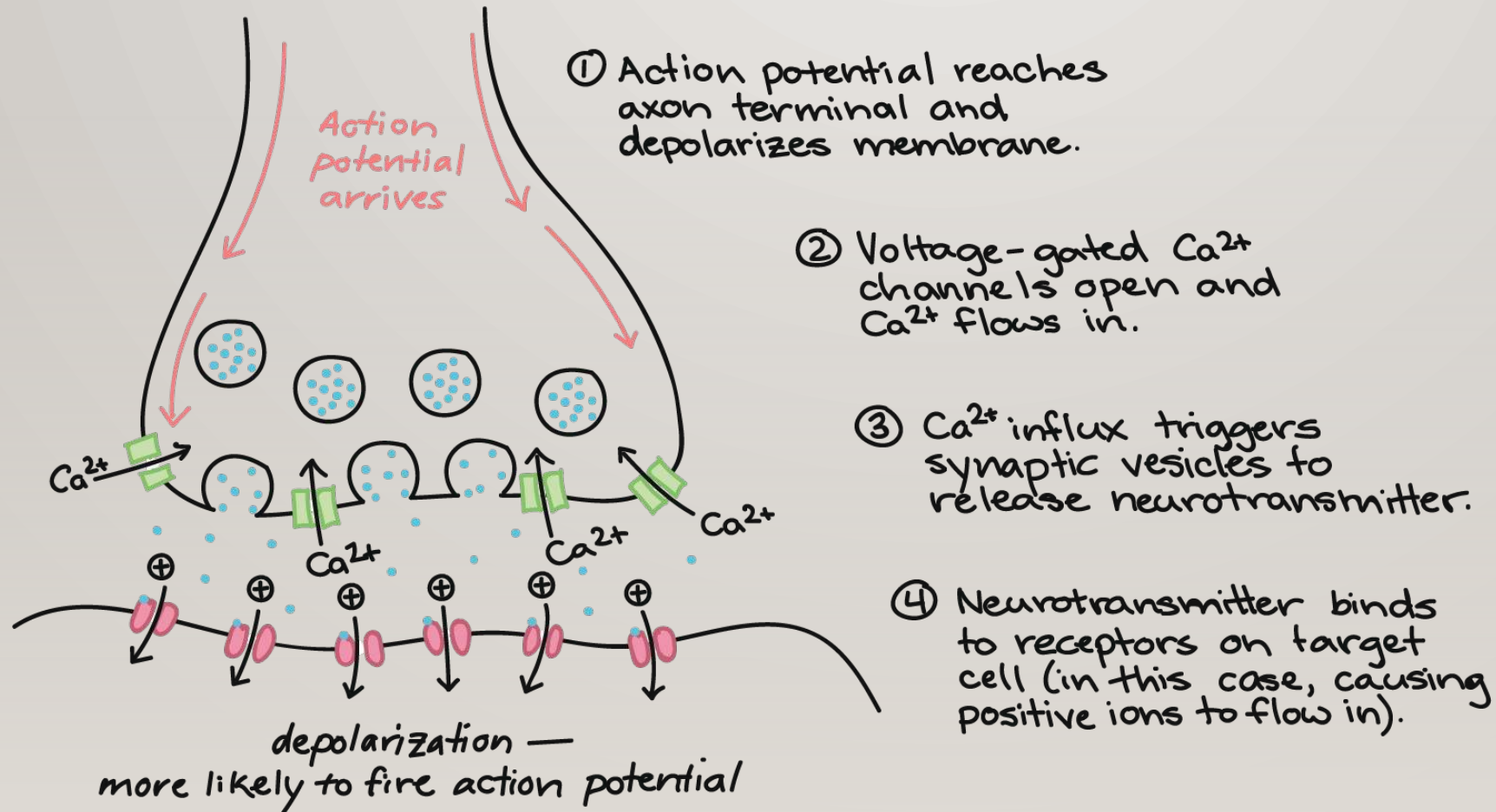
Synapse

Note: We'll learn the details of an action potential next week.

Depolarization meant electrical signal arrives....

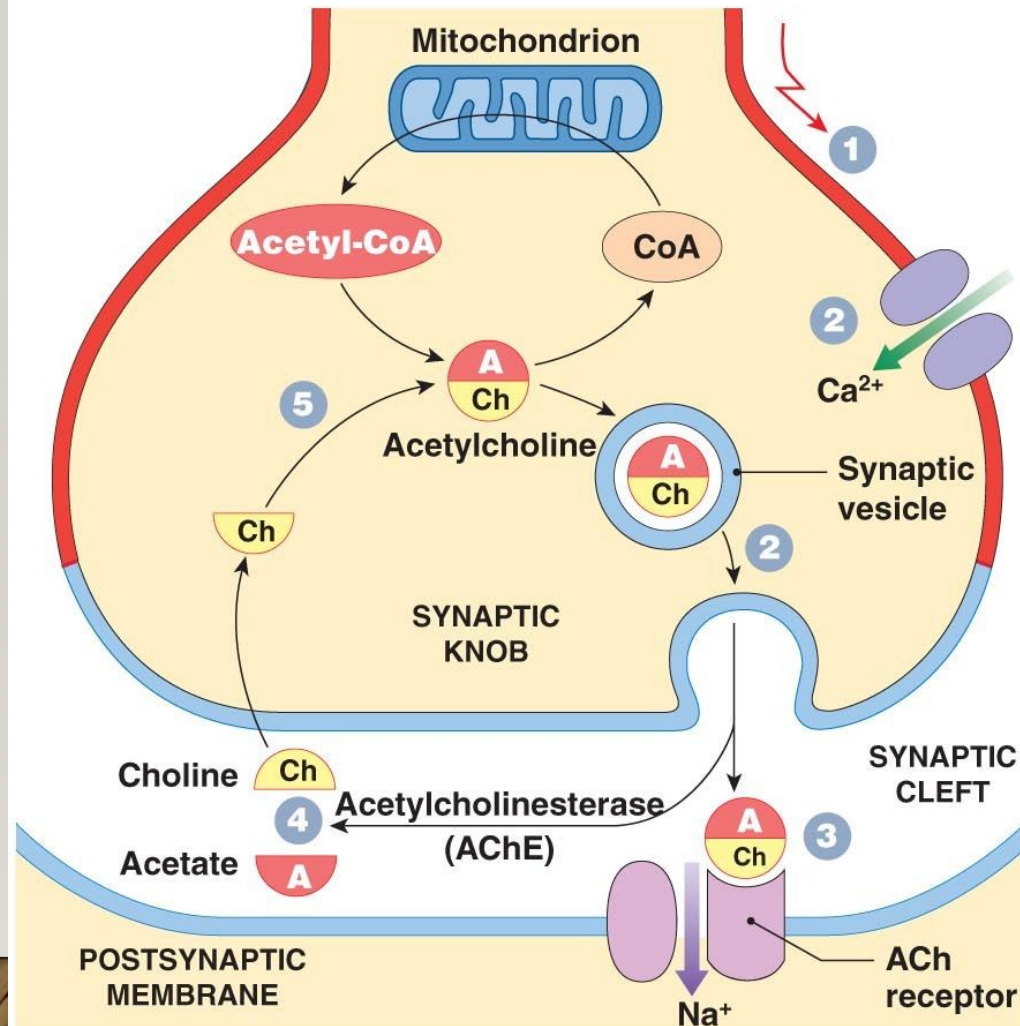
Excitatory
-causes an action potential in receiving tissue.
Eg: Ach

Inhibitory
-decreases/stops an action potential in receiving tissue.
Eg. GABA



CHOLINERGIC SYNAPSE SKELETAL MUSCLES.

The events that occur at a cholinergic synapse



Events Occurring at Synapse

- 1 An arriving action potential depolarizes the synaptic knob.
- 2 Calcium ions enter the cytoplasm, and after a brief delay, ACh is released through the exocytosis of synaptic vesicles.
- 3 ACh binds to sodium channel receptors on the postsynaptic membrane, producing a graded depolarization.
- 4 Depolarization ends as ACh is broken down into acetate and choline by AChE.
- 5 The synaptic knob reabsorbs choline from the synaptic cleft and uses it to synthesize new molecules of ACh.

NEUROTRANSMITTERS

MATCHING -

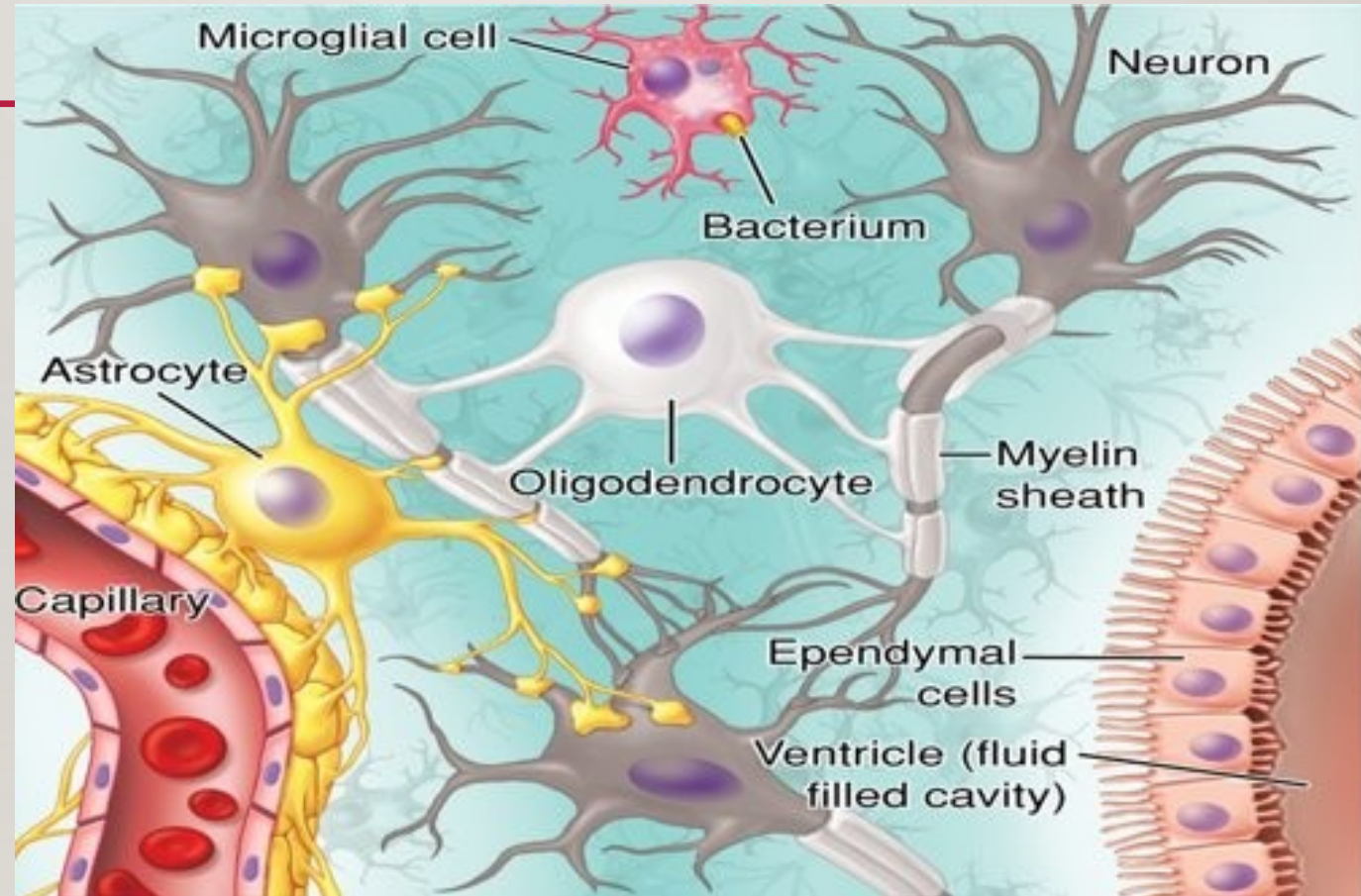
-
- A – Brains ‘reward’ pathway and influences mood/emotional state.
 - B- Used in a cholinergic synapse – skeletal muscles !
 - C- Stimulates vasodilation
 - D- Used in an adrenergic synapse/adrenaline – for sympathetic responses
 - 1-Norepinephrine
 - 2-Acetylcholine
 - 3-Dopamine
 - 4- Nitric oxide

CNS NEUROGLIA – ASTROCYTES

Forms and
regulates the
Blood Brain Barrier

Supports neuron
structure

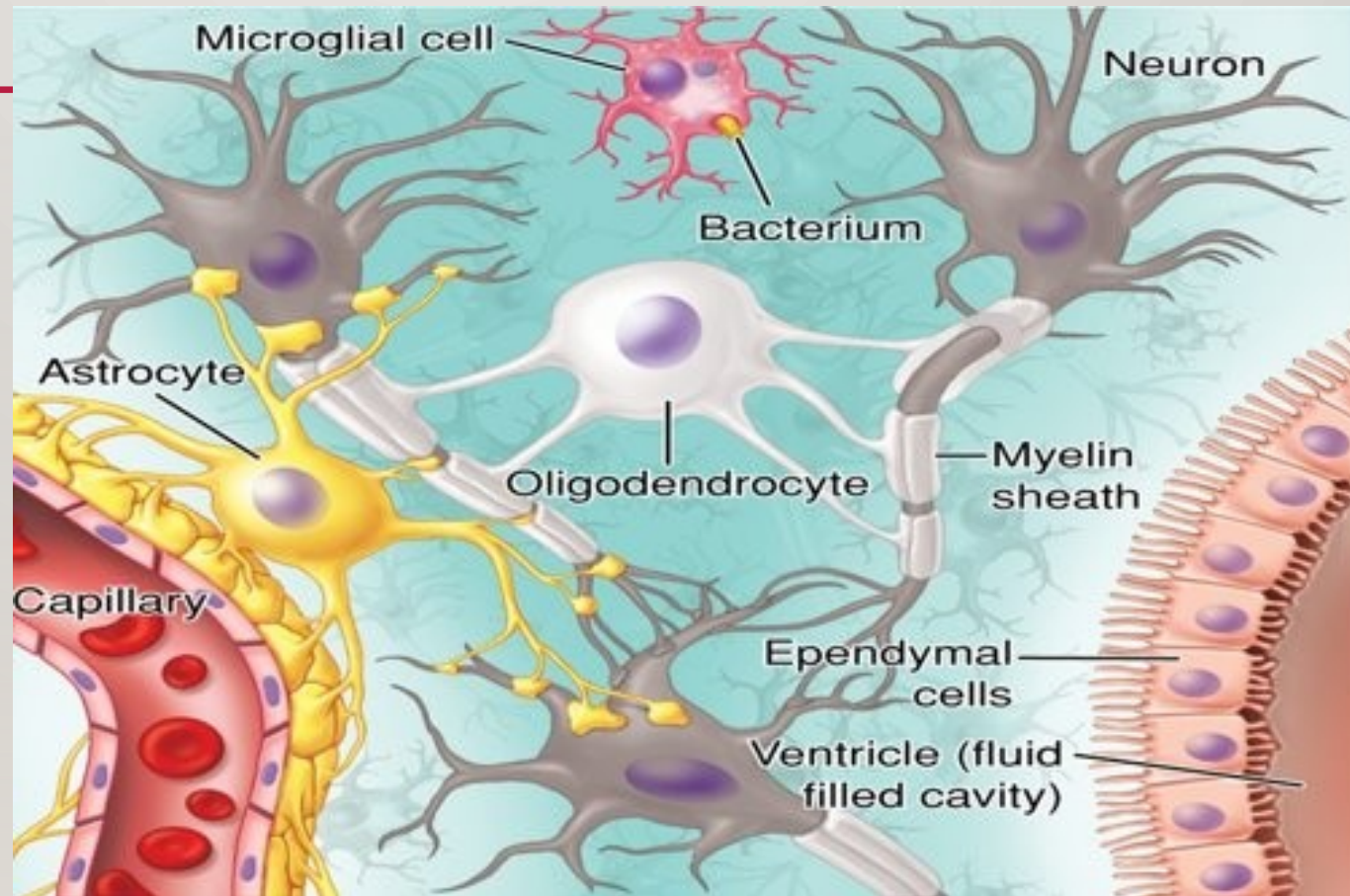
Repairs minor
injuries



CNS NEUROGLIA – OLIGODENDROCYTES

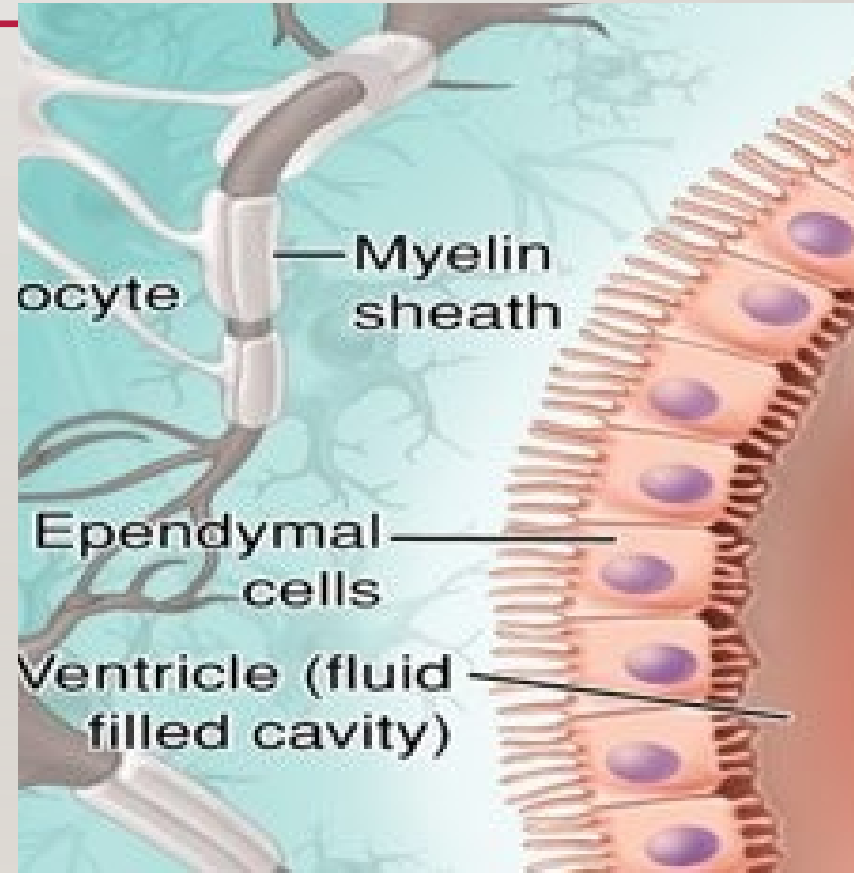
Makes myelin in the
CNS

Myelin speeds up the
conduction of
electrical impulse
along the nerve's
axon – faster
communication.



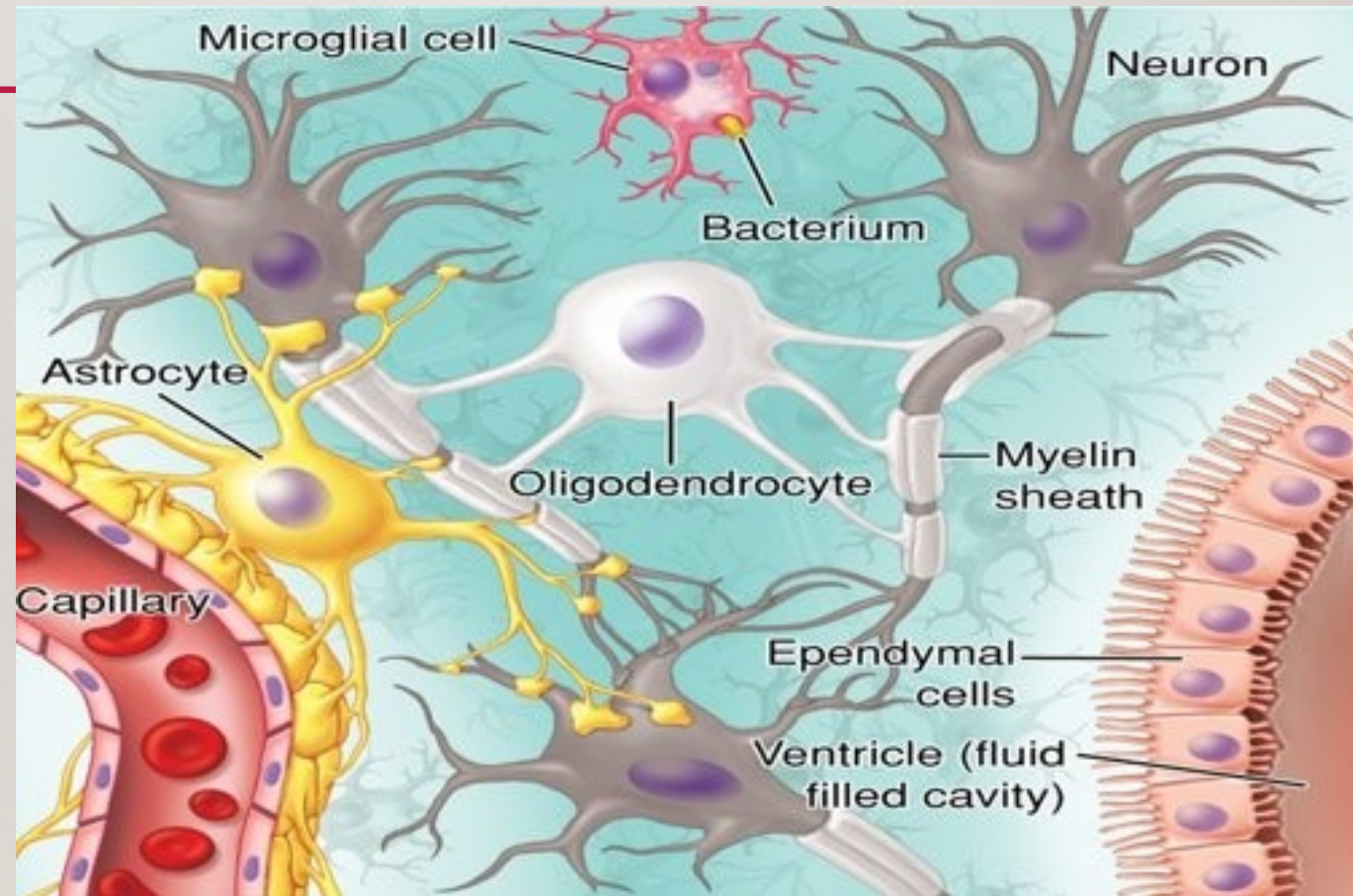
CNS NEUROGLIA – EPENDYMAL CELLS

“pen” makes ink
(CSF)



CNS NEUROGLIA – MICROGLIA

Cleans up debris -
this includes any
pathogens.
This one is a
phagocyte!



NEUROGLIA- PNS

Schwann cells – myelin sheath
in the peripheral system.

Myelin still has the same
function:
Speeds up electrical
activity, faster
communication.



REMINDERS

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 5 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)

Struggling?
Let's meet - send me an
email and we can arrange
a meeting time!

You can avoid feeling
overwhelmed on Sunday by
completing something each
day - this also leads to better
retention of knowledge.